Simpler classical and faster quantum algorithms for Gibbs partition functions

Paweł Wocjan

IBM Quantum, IBM T.J. Watson Research Center

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Simpler classical and faster quantum algorithms for Gibbs partition functions

Research project conducted by the Quantum Algorithms Group:

- Srinivasan Arunachalam
- Vojtech Havlicek
- Giacomo Nannicini
- Kristan Temme
- Pawel Wocjan

Problem

- We consider the problem of approximating the partition function of a classical Hamiltonian using simulated annealing.
- This requires
  - (a) the computation of a cooling schedule
  - (b) the ability to estimate the expectations of random variables that depend on Gibbs distributions at the inverse temperatures of the cooling schedule.

Accomplishments

- We propose classical and quantum algorithms for these two tasks, achieving two goals:
  - (i) we simplify the seminal work of Štefankovič, Vempala and Vigoda (*J. ACM*, 56(3), 2009), improving their running time and almost matching that of the current classical state of the art;
  - (ii) we quantize our new simple algorithm, improving upon the best known quatum algorithm for computing partition functions of many problems, due to Harrow and Wei (SODA 2020).

Accomplishments

- The proposed quantum algorithm has two advantages over the classical algorithm:
  - it has quadratically faster dependence on the spectral gap of the Markov chains as well as the precision, and
  - it computes a shorter cooling schedule, which matches the length conjectured to be optimal by Štefankovič, Vempala and Vigoda.
- (The algorithm requires a fault-tolerant quantum computer.)

Hamiltonians

- The state space Ω is some finite set.
- A (classical) Hamiltonian H is function that assigns an energy $H(x)$ to each state $x \in \Omega$.
- The possible energies are

$$H(x) \in \{0, 1, \ldots, n\}$$

  where n is a positive integer.

Partition functions

- An inverse temperature β is a non-negative real number.
- The partition function at β is

$$Z(\beta) = \sum_{x \in \Omega} \exp\big(-\beta H(x)\big).$$

- The partition function at inverse temperature $\beta_{\mathsf{min}} = 0$ is

$$Z(\beta_{\mathsf{min}}) = |\Omega|.$$

- What can we say about the complexity of computing the partition function $Z(\beta_{\mathsf{max}})$ at a large inverse temperature β_{max}?

Complexity of partition functions

- In general, it is not possible to efficiently compute the exact value $Z(\beta_{\mathsf{max}})$ for large β_{max} since

$$Z(\beta_{\mathsf{max}}) \approx a_0$$

  a_0 is the number of states $x \in \Omega$ with $H(x) = 0$.

- It is easy to construct Hamiltonians whose ground states correspond to the solutions of some NP-hard problems.

- $\Longrightarrow$ The goal is to estimate $Z(\beta_{\mathsf{max}})$.

Estimating partition functions

- Let $\varepsilon \in (0, 1)$ be the error parameter.
- We want to obtain an estimate $\hat{Z}(\beta_{\mathsf{max}})$ of $Z(\beta_{\mathsf{max}})$ such that

$$(1 - \varepsilon) \cdot Z(\beta_{\mathsf{max}}) \ \le \ \hat{Z}(\beta_{\mathsf{max}}) \ \le \ (1 + \varepsilon) \cdot Z(\beta_{\mathsf{max}})$$

holds with high probability ($\ge \frac{3}{4}$).

Estimating partition functions

- Equivalently, the above condition is expressed as

$$|Z(\beta_{\mathsf{max}}) - \hat{Z}(\beta_{\mathsf{max}})| \leq \varepsilon \cdot Z(\beta_{\mathsf{max}})$$

- We say that $\hat{Z}(\beta_{\mathsf{max}})$ estimates $Z(\beta_{\mathsf{max}})$ with relative error ε.
- It is not enough to obtain an estimate with additive error.

Methods and Tools

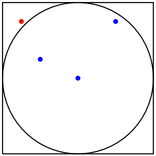
- The algorithms rely on several methods and tools.
- These methods and tools can be logically organized into:
  - Monte Carlo method
    (classical and quantum Chebyshev inequalities)
  - Simulated annealing
    (Gibbs distribution, cooling schedule, product estimators)
  - Markov chains (quantization of Markov chains)

Monte Carlo method

- The Monte Carlo method refers to a collection of tools for estimating values through sampling and simulation.

Monte Carlo method

- Chose a point uniformly at random inside the square with corners $(\pm 1, \pm 1)$.
- The probability of the point landing inside the circle with center point $(0, 0)$ and radius 1 with probability is $\frac{\pi}{4}$.



- $\Longrightarrow$ We can estimate the value of the constant π by:
  - Sampling points uniformly at random from the square
  - Outputting the fraction of points inside the circle

Monte Carlo method

- Let $D : \Omega \to [0, 1]$ be some probability distribution.
- Let $f : \Omega \to \mathbb{R}_{\geq 0}$ be some function.
- The goal is to estimate the expectation

$$\mu = \sum_{x \in \Omega} D(x) \cdot f(x)$$

- Draw m independent samples $x_1, \dots, x_m$ from D
- Output the empirical mean

$$\hat{\mu} = \frac{1}{m} \sum_{x \in \Omega} f(x_i)$$

Monte Carlo method

- Define the second moment

$$\phi = \sum_{x \in \Omega} D(x) \cdot f(x)^2$$

- Define the relative variance to be the ratio

$$\frac{\phi}{\mu^2}$$

Classical Chebyshev inequality

- Assume that $\frac{\phi}{\mu^2} \leq B$ and set

$$m = O\left(\frac{B}{\varepsilon^2}\right)$$

- Chebyshev inequality implies that the empirical mean

$$\hat{\mu} = \frac{1}{m}\sum_{i=1}^{m} f(x_i)$$

of the m independent samples $x_1, \ldots, x_m$ from D satisfies

$$|\mu - \hat{\mu}| \leq \varepsilon \cdot \mu$$

with probability $\geq \frac{3}{4}$.

Quantum samples

- For a probability distribution D, define the corresponding quantum sample $|\psi_D\rangle$

$$|\psi_D\rangle = \sum_{x \in \Omega} \sqrt{D(x)} |x\rangle$$

- Define the reflection R_D around the quantum sample $|\psi_D\rangle$

$$R_D = 2|\psi_D\rangle\langle\psi_D| - I$$

Quantum Chebyshev inequality

- The quantum algorithm achieves a square root speed-up:
  - prepares a constant number of quantum samples $|\psi_D\rangle$
  - invokes the reflection R_D

$$O\left(\frac{\sqrt{B}}{\varepsilon}\right)$$

    many times
  - performs some other transformations (cost is smaller)

- The number of samples for the classical algorithm is

$$O\left(\frac{B}{\varepsilon^2}\right)$$

Intuition behind quantum Chebyshev inequality

- Assume that values $f(x)$ are confined to the interval $[0, 1]$.
- Consider the quantum state $|\psi_{D,f}\rangle$ and the projector P:

$$|\psi_{D,f}\rangle = \sum_{x\in\Omega} \sqrt{D(x)}|x\rangle \otimes \left(\sqrt{f(x)}|1\rangle + \sqrt{1-f(x)}|0\rangle\right)$$

$$P = I \otimes |1\rangle\langle 1|$$

- The desired mean $\mu = \sum_{x\in\Omega} D(x)\cdot f(x)$ is equal to

$$\mu = \langle\psi|P|\psi\rangle$$

- $\Longrightarrow$ We can use amplitude estimation in this special case.
- We handle general f by decomposing f as a weighted sum of functions with bounded output.

Gibbs distribution

- The partition function $Z(\beta)$ at inverse temperature is

$$Z(\beta) = \sum_{x \in \Omega} \exp\big(-\beta H(x)\big)$$

- The Gibbs distribution π_β at β is a distribution on Ω: each state $x \in \Omega$ occurs with probability

$$\pi_\beta(x) = \frac{\exp\big(-\beta H(x)\big)}{Z(\beta)}$$

Estimating ratios of partition functions

- Let β and β' be two inverse temperatures.
- Let D be the Gibbs distribution π_β.
- Let f be the function defined by

$$f(x) = \exp\big(-(\beta' - \beta)H(x)\big)$$

- The corresponding expectation μ is equal to

$$\mu = \sum_{x \in \Omega} D(x) \cdot f(x) = \frac{Z(\beta')}{Z(\beta)}.$$

- $\Longrightarrow$ We can estimate ratios of partition functions provided that we can sample from Gibbs distributions.

Cooling schedule, telescoping product

- We need to determine a suitable cooling schedule $0 = \beta_0 < \beta_1 < \ldots < \beta_\ell = \beta_{\mathsf{max}}$.
- We can now express $Z(\beta_{\mathsf{max}})$ using the following telescoping product:

$$Z(\beta_{\mathsf{max}}) = Z(\beta_0) \cdot \frac{Z(\beta_1)}{Z(\beta_0)} \cdot \frac{Z(\beta_2)}{Z(\beta_1)} \cdots \frac{Z(\beta_\ell)}{Z(\beta_{\ell-1})}$$

Previous approach based on product estimators

- To estimate $Z(\beta_{\text{max}})$, we
  - estimate all ratios
    $$\frac{Z(\beta_{i+1})}{Z(\beta_i)}$$
    with relative error ε/ℓ using quantum Chebyshev
  - take the product of the estimators.
- This can be done efficiently provided that the relative variances at all stages of the cooling schedule are small.

New approach based on paired-product estimators

- Let β_i and β_{i+1} be two adjacent inverse temperatures.
- Let m_i be the midpoint between β_i and β_{i+1}.
- Observe that the ratio $Z(\beta_{i+1})/Z(\beta_i)$ can be expressed as

$$\frac{Z(\beta_{i+1})}{Z(m_i)} \cdot \left(\frac{Z(\beta_i)}{Z(m_i)}\right)^{-1}$$

New approach based on paired-product estimators

- We individually estimate

$$\frac{Z(\beta_{i+1})}{Z(m_i)} \qquad \text{and} \qquad \frac{Z(\beta_i)}{Z(m_i)}$$

- We output their ratio as an estimate for $Z(\beta_{i+1})/Z(\beta_i)$.
- This is highly beneficial because the resulting relative variances are significantly smaller $\Rightarrow$ fewer samples are required.

New approach based on paired-product estimators

- Moreover, the relative resulting variances depend on the overlap

$$\langle \pi_{\beta_i} | \pi_{\beta_{i+1}} \rangle$$

  between the quantum samples of the Gibbs distributions π_{β_i} and $\pi_{\beta_{i+1}}$.

- This overlap can be estimated efficiently on a quantum computer.

- $\Longrightarrow$ We obtain a simple and efficient quantum algorithm for computing well-balanced cooling schedules.

Markov chains

- Markov chains make it possible to sample from complicated high-dimensional probability distributions.
- For instance, the Metropolis algorithm is a method for constructing Markov chains that enable us to sample from Gibbs distributions.

Markov chains

- A Markov chain is described by a stochastic matrix $P = (p_{yx})$.
- Let $x \in \Omega$ be the current state.
- The transition probability p_{yx} is the probability that the next state is $y \in \Omega$.

Mixing times of Markov chains

- P has a limiting distribution π.
- π is the unique eigenvector corresponding to the largest eigenvalue $\lambda_1 = 1$ of P.
- The spectral gap Δ of P is

  $$\Delta = 1 - \lambda_2(P)$$

  where λ_2 denotes second largest eigenvalue of P.
- To obtain one sample from π, it suffices to perform

  $$O\left(\frac{1}{\Delta}\right)$$

  transitions (independent of the initial state).

Quantum Markov chains

- We can construct a quantum Markov chain U.
- U is a unitary such that
  - its unique eigenvector with eigenvalue 1 (eigenphase 0) is the quantum sample
    $$|\psi_\pi\rangle = \sum_{x\in\Omega} \sqrt{\pi(x)}|x\rangle$$
  - all other eigenvectors have eigenphases that are at least
    $$\frac{1}{\sqrt{\Delta}}$$
    away from 0.

Quantum Markov chains

- We can implement the reflection

$$R_\pi = 2|\psi_\pi\rangle\langle\psi_\pi| - I$$

by performing quantum phase estimation of U.

- We have to invoke the unitary U

$$O\left(\frac{1}{\sqrt{\Delta}}\right)$$

times to implement R_π.

- We have to invoke the stochastic matrix P

$$O\left(\frac{1}{\Delta}\right)$$

times to obtain a sample from limiting distribution π.